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An mHealth (Mobile Health) Intervention for Smoking Cessation in People With Tuberculosis

A Cluster Randomized Clinical Trial

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IMPORTANCE Smoking worsens outcomes in people with tuberculosis (TB), while quitting hastens recovery.

OBJECTIVE To assess the effectiveness of a mobile health (mHealth) intervention for achieving self-reported continuous tobacco abstinence at 6 months, supported by biochemical verification at 6 months, compared with usual care in people with TB.

DESIGN, SETTING, AND PARTICIPANTS Multicenter, cluster randomized clinical trial conducted between September 18, 2023, and January 2, 2025, randomizing (2:1) 27 TB clinics in Bangladesh and Pakistan to mHealth or usual care groups. The follow-up was 6 months. Participants were eligible if they were 15 years or older, had been diagnosed with drug-sensitive pulmonary TB in the past 4 weeks, smoked daily, were willing to quit, and had access to mobile phones.

INTERVENTION The mHealth group (n = 720) received text messages throughout TB treatment, daily for 2 months then monthly for 4 months, encouraging tobacco cessation. The usual care group (n = 360) received written information on tobacco cessation.

MAIN OUTCOMES AND MEASURES The primary outcome was self-reported continuous abstinence at 6 months, verified biochemically using carbon monoxide breath test at 6 months. Secondary outcomes included self-reported point abstinence at 9 weeks and 6 months, TB treatment adherence (days receiving TB treatment), TB treatment success (cured + completed treatment), TB treatment failure, TB treatment default (interruption of TB treatment for ≥2 months), and death.

RESULTS Of 9232 patients assessed for eligibility, 1080 were randomized; most were male (mHealth, 96.9%; usual care, 95.8%), and 985 were retained throughout the trial (91%). For the primary outcome, 300 of 720 participants (41.7%) in the mHealth group demonstrated self-reported and biochemically verified continuous abstinence at 6 months, compared with 55 of 360 (15.3%) in the usual care group (risk ratio, 3.0 [95% CI, 2.0-4.9]). In the mHealth vs usual care groups, respectively, mean TB treatment adherence was 174.3 (SD, 21.5) days vs 178.0 (SD, 12.1) days ($P = .23$), and treatment success was 89.3% vs 85.6% (risk ratio, 1.2 [95% CI, 0.9-1.6]). TB treatment failure (0.1% vs 0.5%) and default (3.1% vs 1.9%) were uncommon. Mortality was lower with mHealth (3.5%) vs usual care (7.5%) (hazard ratio, 0.4 [95% CI, 0.2-0.9]).

CONCLUSIONS AND RELEVANCE An mHealth intervention was effective in achieving continuous abstinence in people with TB who smoked. mHealth is a feasible and effective intervention to help patients with TB quit smoking.

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Tobacco consumption poses a substantial threat to global health, with more than 8.7 million deaths, a loss of 230 million disability-adjusted life-years, and a cost of more than US \$1.4 trillion, annually.^{1,2} Despite a reduction in smoking prevalence,³ the persistently high absolute numbers (>1.3 billion) of people who use tobacco means that intensified tobacco control efforts are required. This need is greatest in low- and middle-income countries, where more than 80% of people who use tobacco live⁴ and of whom approximately 90% are men.⁵

Tobacco cessation is central to global efforts to reduce tobacco-related harm. Mobile health (mHealth) interventions for tobacco cessation are promising due to their accessibility and scalability.⁶ While the evidence supporting their effectiveness comes mainly from high-income countries,⁷ they have potential to make tobacco cessation accessible to populations in low- and middle-income countries.⁸ This study focuses on patients with tuberculosis (TB), a neglected group that may benefit from an mHealth intervention.

Approximately 17% to 32% of individuals with TB currently smoke⁹; yet smoking is not adequately addressed for several reasons, eg, lack of capacity, interest, and resources.¹⁰ Due to these barriers,¹¹ no country with high TB burden offers face-to-face tobacco cessation within TB services. Recognizing these challenges, the World Health Organization (WHO) developed an mHealth tobacco cessation intervention that delivers TB-specific text messages to people with TB who smoke.¹² However, high-quality evidence supporting the effectiveness of mHealth tobacco cessation interventions in patients with TB is lacking. The study hypothesis was that relative to usual care, patients with TB receiving an mHealth intervention would be more likely to achieve abstinence at 6 months. This article reports the findings of a cluster randomized clinical trial (RCT), in Bangladesh and Pakistan (2 countries with high burden of tobacco and TB), which tested this hypothesis.

Methods

Trial Oversight

A trial steering committee met annually and reviewed the trial protocol, questionnaires, patient information sheet, consent forms, trial procedures, and progress. Ethics approvals were obtained from the respective committees at the University of Edinburgh, UK, and authorized national bodies in Bangladesh and Pakistan. The trial complied with the UK General Data Protection Regulation and the Data Protection Act 2018.

Trial Design and Participants

We conducted a 2-group, multicenter, cluster RCT in Bangladesh (Dhaka) and Pakistan (Punjab) from September 18, 2023, to January 2, 2025. A cluster design was chosen to minimize contamination and because it was ethically simpler to implement. Twenty-seven TB clinics (clusters) were randomized, in a 2:1 ratio, to the mHealth or usual care groups. In the published protocol, this trial is referred to as the third phase of the project.¹³

Key Points

Question What is the effectiveness of a mHealth (mobile health) intervention for achieving self-reported continuous tobacco abstinence at 6 months, compared with usual care in people with tuberculosis (TB)?

Findings In this randomized clinical trial that included 1080 patients with TB who smoked tobacco, the proportion achieving self-reported continuous tobacco abstinence at 6 months, supported by biochemical verification at 6 months, was significantly higher with mHealth (41.7%) vs with usual care (15.3%).

Meaning In people with TB who smoke tobacco, the use of an mHealth intervention may be warranted.

Government-approved TB clinics, registering at least 50 new patients with TB per month, were eligible. Patients 15 years or older, diagnosed with drug-sensitive pulmonary TB in the last 4 weeks, people who smoked daily (at least 1 puff on ≥ 25 days in the last month), willing to quit and having access to mobile phones, were eligible. Patients treated for multidrug resistance, miliary or extrapulmonary TB, or for tobacco dependence were excluded.

Trial Procedures

An independent statistician, blinded to TB clinics, used computer-generated random-number lists to allocate clusters in a 2:1 ratio to the mHealth and usual care groups. With more clusters offering mHealth, the trial was expected to recruit faster than with equal allocation.¹⁴ Randomization was stratified by country. While the nature of the intervention (mHealth) and the cluster design precluded masking the participants and researchers, the statistician remained masked during the analysis.

At the TB clinics, health care staff identified patients newly diagnosed with TB and referred them to the researchers for eligibility assessment. Those eligible provided written informed consent; for those younger than 18 years, parents/guardians provided written consent. Data were collected on day 0, week 9, and month 6 from all participants, corresponding with routine TB clinic visits. If a participant missed their appointment, they were assessed within 5 days, via telephone or home visit.

To monitor intervention fidelity in the mHealth group, participants were called weekly to check if they were receiving and reading the messages or having the messages read to them. Those in the control group were also called weekly, to detect any contamination. A custom-made mobile app was used to collect the data, which were managed using a secure web platform.

Trial Intervention

The mHealth intervention was designed to deliver TB-specific text messages to people with TB.¹² These messages, together with a guide, were developed by a group of public health experts, epidemiologists, and behavioral scientists convened by WHO in 2019.¹² The messages were grounded in the behavior

change technique taxonomy, a classification system of theory-informed techniques used to modify behavior.¹⁵ The messages were translated into Bangla and Urdu using WHO's guidance,¹⁶ then reviewed and refined with feedback from a patient group (phase 1). A pilot study involving 16 patients with TB who smoked assessed user experiences with clarity, quantity, timing, and frequency of messages (phase 2). Following trial enrollment, participants first received 3 welcoming messages explaining the program's purpose and schedule. While the program was not interactive, participants could opt out of receiving text messages by calling or texting the researchers.

The mHealth text messages supported cessation with motivational messages and coping strategies to reduce cravings and address withdrawal symptoms. Messages to promote healthy behaviors were also included. Alongside emphasizing the importance of taking TB medications and follow-up appointments, text messages addressed the association between TB and tobacco use, highlighting how smoking slows recovery and increases the risk of relapse. In total, 134 unique text messages were sent through a web-based application to participants' mobile numbers over 6 months. One hundred messages were delivered (3-4 per day) in the first month, followed by 30 messages (1 per day) in the second, and then 4 messages (1 per month) over the next 4 months. The last 4 messages were reminders for attending TB clinics and collecting medications. The system reported weekly on the number of sent messages and their recipients.

Participants in both groups received TB treatment consisting of rifampicin, isoniazid, pyrazinamide, and ethambutol. After 2 months, the regimen was reduced to rifampicin and isoniazid. In addition, written information on tobacco-related harms and quit advice were also offered to both groups. None were offered medications to help them quit because medications were neither available in the TB clinics nor accessible privately due to their prohibitive cost.

Trial Outcomes

The primary outcome was biochemically verified continuous abstinence at 6 months after randomization. Abstinence was defined as self-report of not having used more than 5 cigarettes, bidis, or water pipe sessions since the quit date (set 5 days after enrollment), verified biochemically by a breath carbon monoxide reading of less than 10 ppm at month 6.¹⁷ In the case of concomitant smokeless tobacco use at baseline, a rapid test cassette (Hangzhou AllTest Biotech Co Ltd) was used to detect cotinine (a nicotine metabolite) in urine samples. A negative result verified abstinence.

We performed exploratory analyses of the following secondary outcomes: point abstinence, self-reported abstinence from tobacco in the previous 7 days, assessed at week 9 and month 6, adherence to TB treatment (number of days TB medication was taken), TB success rate (cured + treatment completion), TB treatment failure, TB treatment defaults (interruption of TB treatment for ≥ 2 months) and deaths at month 6. Outcome details are provided in the trial protocol in [Supplement 1](#) and in the published protocol.¹³

The sample size (phase 3 in the protocol) was estimated using 90% power, 5% significance level (2-sided), and absti-

nence probabilities of 18% and 8% for the mHealth and usual care groups, respectively, at 6 months.¹⁸ An intracluster correlation coefficient of 0.02 was used based on the recommendations for behavioral outcomes in cluster RCTs.¹⁹ For a cluster size of 26 participants, we estimated a design effect of 1.5. After adjusting for this and an anticipated 20% attrition rate, we arrived at a sample size of 1080 participants and 27 clusters (capped at 40 participants per cluster).

Statistical Analysis

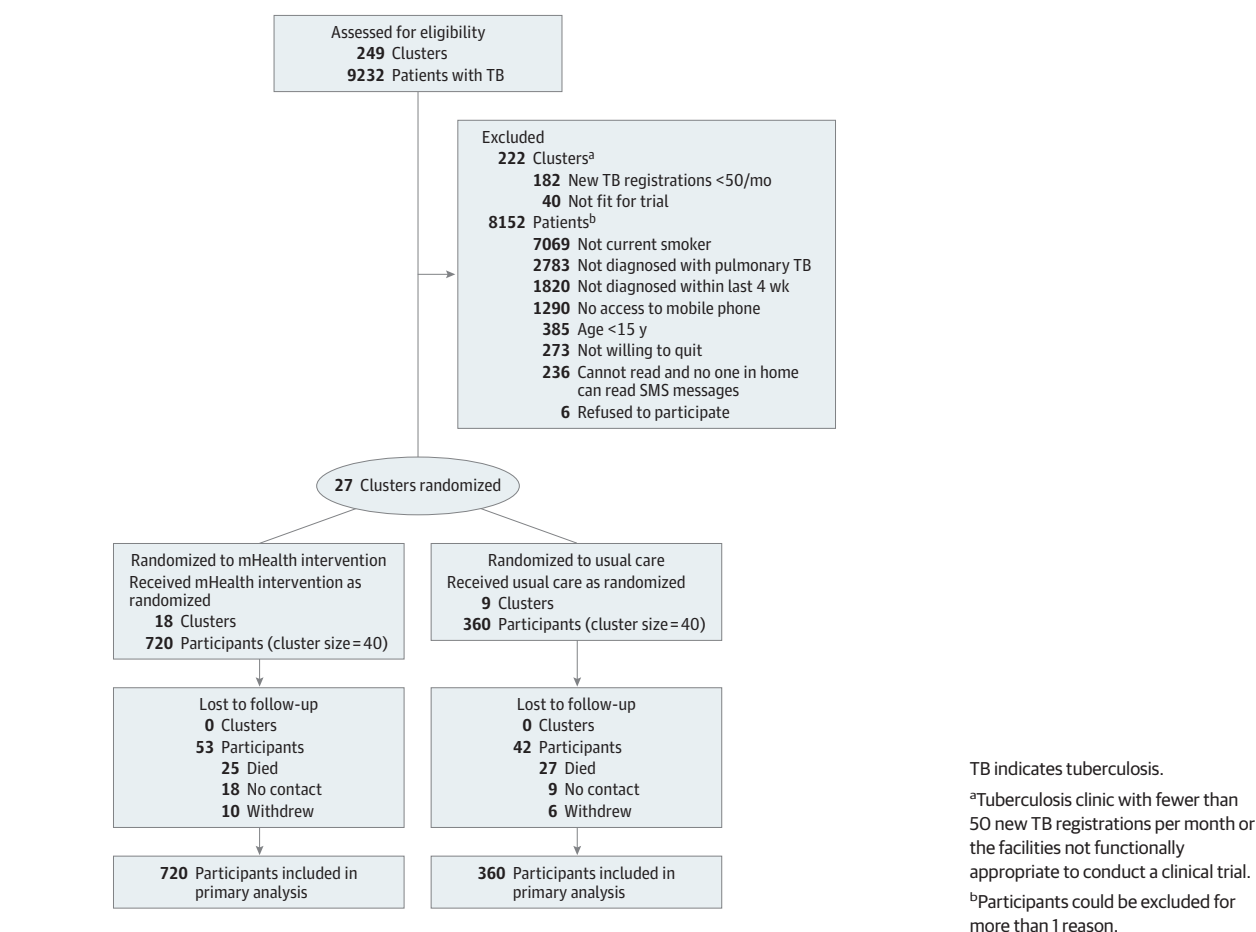
We reported the analyses in accordance with CONSORT guidelines (extension for cluster RCT).²⁰ R version 4.4.3 (R Foundation) and JASP Team (2024; version 0.19.3) were used for data analysis.²¹

Data normality was tested visually and by using the Shapiro-Wilk test. Categorical outcomes were summarized using frequencies and percentages and continuous outcomes using means and standard deviations. A mixed-effects model was used to assess the effect of mHealth on abstinence outcomes, incorporating fixed effects for the intervention and relevant covariates selected post hoc (age, sex, education, occupation, and smoking duration) and random effects for clusters. Logistic regression within the mixed-effects framework was used to analyze categorical outcomes, accounting for cluster-level variability and intervention effects. Additionally, a bayesian hierarchical logistic regression model was fit with the *brms* package in R to estimate cluster-level quit probabilities and 95% credible intervals. We overlaid observed proportions to compare raw data with our model-based results. For all analyses, coefficients of intraclass correlation were estimated from the variance components of a mixed-effects logistic regression model using the *lme4* package in R. For comparing death rates between the 2 groups over 6 months, we used the Cox proportional hazards frailty model, adjusting for site-level clustering. The secondary outcomes were exploratory, and the analysis was not adjusted for multiple comparisons. A post hoc sensitivity analysis using a carbon monoxide cutoff of less than 6 ppm was performed for the primary outcome.

Consistent with standard practice in smoking cessation trials, missing primary outcome data were treated as a negative outcome (ie, not abstinent). The intervention effect estimates were assessed through both intention-to-treat and complete case analyses (those with missing primary outcome data were discarded).²² A post hoc sensitivity analysis was also carried out after excluding deaths.

We recorded the number and reasons for ineligibility and calculated cluster-level recruitment, abstinence, TB treatment success, and death rates. Using multilevel modeling, we also explored associations between cluster characteristics and the primary outcome. We adjusted for relevant sociodemographic covariates (selected post hoc) in the primary analysis. We compared the primary outcome between the following prespecified subgroups: age (<40 years and ≥ 40 years); education (no formal education, primary, secondary or higher); employment (active, dependent, retired); and smoking duration (<24 years and ≥ 24 years). All statistical tests were 2-sided, and only *P* values less than .05 were considered statistically significant.

Figure 1. Flow of a Trial of a Mobile Health Intervention for Smoking Cessation in People With Tuberculosis



Results

Participants

We assessed 9232 patients for eligibility and enrolled 1080 (mHealth, 720; usual care, 360) participants (Figure 1). Six patients refused to participate; the remainder (8146) did not meet inclusion criteria (eTable 1 in Supplement 2); the main reasons for exclusion were not smoking, nonpulmonary TB, and TB diagnosis more than 4 weeks previously.

Baseline characteristics of the clusters and participants are available in Table 1. Compared with the usual care group, the mHealth group had more people who smoked cigarettes (94.3% vs 90.6%) and used smokeless tobacco (6.9% vs 2.2%). The mHealth group consumed a lower average number of cigarettes per day (5 vs 7) and bidis per day (8 vs 15) than the usual care group. More participants in the mHealth than the usual care group made previous quit attempts (24.7% vs 9.2%). The imbalance in prior quit attempts across groups was driven by relatively few intervention clusters; eTable 2 in Supplement 2 shows the site-level distribution of prior quit attempts.

All participants in the mHealth group confirmed receiving text messages. At month 6, 8.8% of participants did not provide a primary outcome: 4.8% had died, 1.5% withdrew, and

2.5% lost contact. Except for deaths, attrition rates were comparable between groups (Figure 1). There were no missing data on outcomes, other than for these participants. No participant reported using medications for tobacco cessation. There were also no incidents of contamination based on system-generated reports and calls to participants.

Primary Outcome

Based on self-report only, 47.5% of participants remained abstinent in the mHealth group, compared with 19.4% in the usual care group. On biochemical verification (carbon monoxide [CO] cutoff <10 ppm) and as per intention-to-treat analysis, 41.7% (95% CI, 38.0%-45.4%) of participants demonstrated continuous abstinence in the mHealth group, compared with 15.3% (95% CI, 11.7%-19.4%) in the usual care group (risk ratio [RR], 3.0 [95% CI, 2.0-4.9]) (Table 2). The association persisted after adjustments for baseline covariates selected post hoc (age, sex, education, occupation, and smoking duration) (adjusted risk ratio, 3.2 [95% CI, 2.2-5.2]). In those who achieved biochemically verified continuous abstinence, the mean CO levels (mHealth, 3.54 [SD, 2.1] ppm; usual care, 4.38 [SD, 2.8] ppm) were similar in both groups. The post hoc analysis using a CO cutoff of less than 6 ppm was also consistent with the primary analysis (RR, 3.7 [95% CI, 2.4-5.8]), confirming the

Table 1. Baseline Characteristics of the Study Groups

Characteristic	Group	
	mHealth	Usual care
Clusters, No.		
Total	18	9
Cluster size	40	40
Rural setting	11	7
Average No. of TB cases/mo	142	93
Participants		
Total No.	720	360
Age, mean (SD), y	48.7 (15.8)	48.7 (15.9)
Sex, No. (%)		
Male	698 (96.9)	345 (95.8)
Female	22 (3.1)	15 (4.2)
BMI, median (IQR) ^a	18.7 (4)	18.6 (4)
Education, No. (%)		
No formal education	315 (43.8)	155 (43.1)
Primary	175 (24.3)	114 (31.7)
Middle	109 (15.1)	43 (11.9)
Secondary	89 (12.4)	36 (10)
Higher	32 (4.4)	12 (3.3)
Marital status, No. (%)		
Single	83 (11.5)	37 (10.3)
Married	607 (84.3)	304 (84.4)
Separated/divorced	30 (4.2)	19 (5.3)
Employment status		
Employed	560 (77.8)	275 (76.4)
Dependents ^b	128 (17.8)	71 (19.7)
Retired	32 (4.4)	14 (3.9)
Able to read messages, No. (%)	425 (59.0)	225 (62.5)
TB stage, No. (%) ^c		
1	629 (87.4)	326 (90.6)
2	45 (6.3)	20 (5.6)
3	46 (6.4)	14 (3.9)
Smoking type, No. (%)		
Cigarettes	679 (94.3)	326 (90.6)
Hookah (water pipe)	61 (8.5)	35 (9.7)
Bidi (hand-rolled)	21 (2.9)	21 (5.8)
Concurrent smokeless tobacco use	50 (6.9)	8 (2.2)
Tobacco use/d, median (IQR)		
Bidi (hand-rolled)	8 (10.0)	15 (14.0)
Cigarettes	5 (6.0)	7 (6.0)
Hookah (water pipe)	3 (4.0)	3 (2.0)
Smoking allowed inside homes, No. (%)	482 (66.9)	189 (52.5)
Attempted to quit in past, No. (%)	178 (24.7)	33 (9.2)
Mean smoking duration (SD), y	24.8 (15.1)	24.6 (15.2)
Median smoking start age (IQR), y	20 (9.0)	20 (8.0)
Heaviness of Smoking Index, ²³ low addiction, No. (%)	514 (75.7)	212 (65.0)

Abbreviations: BMI, body mass index; TB, tuberculosis.

^a Calculated as weight in kilograms divided by square of the height in meters.

^b Unemployed, homemakers, students, or otherwise not in active employment.

^c Clinical severity based on the number of TB signs and symptoms: stage 1, mild to moderate TB disease; stage 2, severe TB disease; stage 3, very severe TB disease.

robustness of the intervention effect. The bayesian hierarchical model (Figure 2) showed heterogeneity of probabilities of quitting among clusters. Overall, clusters in the mHealth group had higher fitted rates than those in the usual care group.

Secondary Outcomes

The self-reported point abstinence at week 9 and month 6 was also higher in the mHealth group than the usual care group (Table 2). Mean treatment adherence was similar between groups (174.3 [SD, 21.5] days for mHealth vs 178.0 [SD, 12.1] days for usual care; $P = .23$) (eTable 3 in Supplement 2). Treatment success occurred in 643 of 720 mHealth participants (89.3%) and 308 of 360 usual care participants (85.6%) (RR, 1.2 [95% CI, 0.9-1.6]). Treatment failure was rare (0.1% [1/720] in the mHealth group vs 0.5% [2/360] in the usual care group), and treatment default occurred in 3.1% (22/720) vs 1.9% (7/360) participants, respectively (Table 2). There were fewer deaths in the mHealth group than in the usual care group (3.5% vs 7.5%; hazard ratio, 0.4 [95% CI, 0.2-0.9]) (Table 2). Tuberculosis disease (61.5% [32/52]) was the most common cause of death (eTable 4 in Supplement 2). The Kaplan-Meier plot showed higher probability of survival over time in the mHealth group compared with the usual care group (eFigure in Supplement 2).

All clusters were of similar size; however, variations were observed in the recruitment, abstinence, successful TB treatment, and death rates (eTable 5 and eTable 6 in Supplement 2). The main cluster characteristics that modified the primary outcome included shorter smoking duration and higher educational status (eTable 7 in Supplement 2). Subgroup analyses demonstrated that the treatment effect remained consistent across age, education, employment, ability to read Short Message Service messages, and smoking duration categories (eTable 8 in Supplement 2). Results of post hoc sensitivity analysis after excluding deaths were also consistent with those from the primary intention-to-treat analysis (eTable 9 in Supplement 2).

Adverse Events

Minor events reported included nausea (23.0% for mHealth vs 22.3% for usual care), diarrhea (7.5% vs 7.5%), dry mouth (62.7% vs 55.7%), epigastric pain (27.7% vs 40.4%), headache (45.1% vs 49.1%), insomnia (35.3% vs 33.5%), abnormal dreams (10% vs 13.2%), irritability (40.5% vs 43.4%), anxiety (33.3% vs 36.8%), palpitations (31% vs 28.4%), and musculoskeletal pain (61.4% vs 60.8%). Of these, dry mouth, irritability, and anxiety were more common in the mHealth group (eTable 10 in Supplement 2).

Discussion

In people with TB who smoke, an mHealth intervention led to increased self-reported continuous abstinence supported by biochemical verification at month 6 compared with usual care. A similar difference was observed between groups for self-reported point abstinence at week 9 and month 6. While TB treatment adherence and TB success rates did not differ

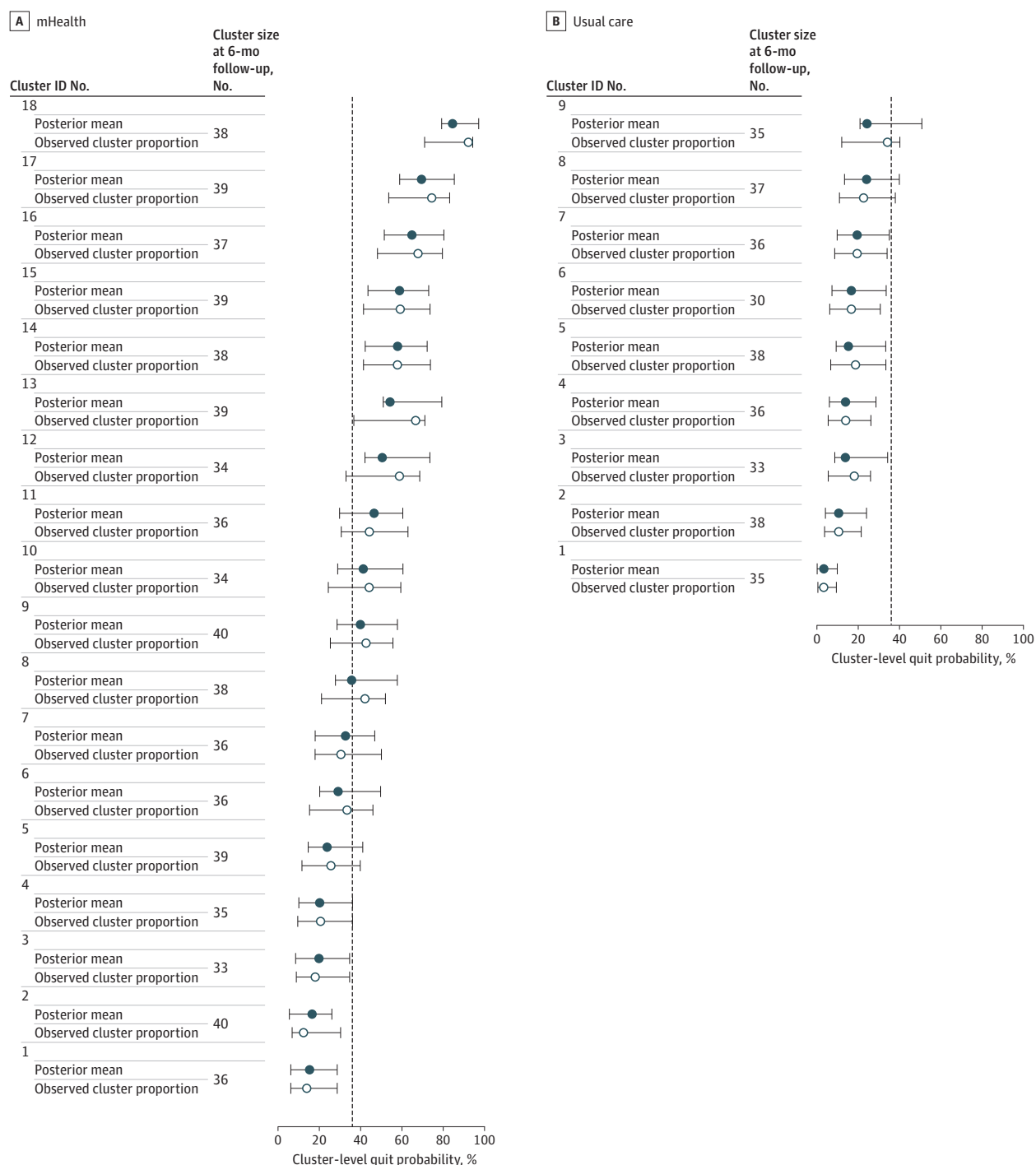
Table 2. Comparison of Primary and Secondary Outcomes Between Study Groups

Outcomes	mHealth		Usual care		Absolute difference (95% CI)		Crude		Adjusted ^a	
	No./No. ^b	% (95% CI)	No./No. ^b	% (95% CI)			RR (95% CI)	ICC ^c	RR/HR (95% CI) ^{c,d}	ICC
Primary outcome										
Biochemically verified abstinence at mo 6, <10 ppm (ITT) ^e	300/720	41.7 (38.0 to 45.4)	55/360	15.3 (11.7 to 19.4)	26.4 (21.0 to 31.6)		3.0 (2.0 to 4.9)	0.18	3.2 (2.2 to 5.2)	0.18
Biochemically verified abstinence at mo 6, <10 ppm (PP) ^e	300/667	45 (41.2 to 48.8)	55/318	17.3 (13.3 to 21.9)	27.7 (22.1 to 33.3)		2.9 (2.0 to 4.7)	0.19	3.1 (2.1 to 5.2)	0.19
Biochemically verified abstinence, <6 ppm (ITT) ^e	264/720	36.7 (33.1 to 40.3)	38/360	10.6 (7.6 to 14.2)	26.1 (21.2 to 30.7)		3.7 (2.4 to 5.8)	0.16	3.9 (2.4 to 6.9)	0.18
Biochemically verified abstinence at mo 6, <6 ppm (PP) ^e	264/667	39.6 (35.9 to 43.4)	38/318	11.9 (8.6 to 16.1)	27.7 (22.5 to 32.8)		3.6 (2.14 to 6.87)	0.17	3.8 (2.3 to 7.7)	0.18
Continuous abstinence, self-reported only (ITT)	342/720	47.5 (43.8 to 51.2)	70/360	19.4 (15.5 to 23.9)	28.1 (22.6 to 33.5)		2.7 (1.8 to 4.1)	0.19	2.8 (1.9 to 4.2)	0.19
Secondary outcomes										
Point abstinence (last 7 d) at wk 9 (ITT)	353/720	49.0 (45.3 to 52.7)	75/360	20.8 (16.7 to 25.4)	28.2 (22.6 to 33.8)		2.6 (1.8 to 3.9)	0.19	2.7 (1.8 to 4.3)	0.18
Point abstinence (last 7 d) at mo 6 (ITT)	400/720	55.6 (51.8 to 59.2)	82/360	22.8 (18.5 to 27.5)	32.8 (27.1 to 38.4)		2.7 (2.0 to 3.8)	0.19	2.7 (1.9 to 4.0)	0.19
TB treatment success (cured + completed treatment)	643/720	89.3 (86.8 to 91.5)	308/360	85.6 (81.5 to 89.0)	3.8 (−0.5 to 8.2)		1.2 (0.9 to 1.6)	0.16	1.2 (0.9 to 1.5)	0.15
Death	25/720	3.5 (2.2 to 5.0)	27/360	7.5 (5.0 to 10.7)	4 (1.0 to 7.1)		NA	NA	0.4 (0.2 to 0.9)	NA
Treatment default (interruption of TB treatment for ≥2 mo) ^f	22/720	3.1 (1.9 to 4.6)	7/360	1.9 (0.8 to 4.0)	1.1 (−0.8 to 3.0)		NA	NA	NA	NA
TB treatment failure ^f	1/720	0.1 (0.01 to 0.8)	2/360	0.5 (0.1 to 2.0)	0.4 (−0.4 to 1.2)		NA	NA	NA	NA

Abbreviations: HR, hazard ratio; ICC, intraclass correlation coefficient; ITT, intention-to-treat analysis; NA, not applicable; PP, per-protocol analysis; RR, relative risk.

^a Adjusted (post hoc) for age, sex, education, occupation, smoking duration: Adjusted analyses accounted for clustering, mixed-effects models for RR and shared frailty Cox models for HR.^b Number of outcomes/total number in the group.^c Intraclass correlation coefficient reported for RR models only.^d Hazard Ratio estimated from a Cox proportional hazards frailty model.^e Carbon-monoxide breath test cut-off values.^f Mixed-effects models could not be estimated for treatment default and treatment failure due to very low numbers of events across clusters.

Figure 2. Cluster-Level Fitted Quit Probabilities From the Bayesian Hierarchical Model for Tuberculosis Clinics



Posterior means include fixed + random effects. Whiskers indicate 95% credible intervals. ID indicates identification; mHealth, mobile health.

between groups, there were fewer deaths in the mHealth than in the usual care group.

While mHealth interventions have been tested in people with TB,²⁴ outcomes have been limited to treatment adherence²⁵ and TB treatment outcomes.^{26,27} In a trial testing the effectiveness of a mHealth intervention on TB treatment outcomes,

no significant difference was reported in smoking abstinence (a secondary outcome) between those receiving automated text messaging and those receiving usual care.²⁸

To our knowledge, this is the first trial assessing the effectiveness of a mHealth intervention in people with TB who smoke in which tobacco abstinence was biochemically verified.

The mHealth text messages were delivered to all participants as intended, and all clusters and more than 90% of participants were retained throughout the trial. The findings are consistent with those from studies of face-to-face behavioral support in achieving tobacco abstinence in patients with TB. A recent systematic review with 14 studies reported that smoking cessation interventions can achieve between 15% and 82% abstinence in patients with TB.²⁹ A mobile-based telecounseling intervention among individuals with TB who smoked achieved higher quit rates than those receiving brief advice at 6 months (67.5% vs 42%; RR, 1.60 [95% CI, 1.19-2.16]).³⁰ A higher effect size was observed for tobacco abstinence in people with TB as compared with the aforementioned review⁷ findings based on 13 studies conducted in the general population, mostly in high-income countries. In prior studies, behavioral support was associated with a higher abstinence rate in people with TB who smoke as compared with the general population.^{31,32} Access to tobacco cessation interventions (such as quit lines, behavioral support, and medications) is virtually nonexistent in both Bangladesh and Pakistan.³³ For most participants, this was likely their first exposure to any cessation intervention, which makes the observed effect more plausible.

In exploratory analyses, 4.8% of the trial participants died of TB within 6 months; a similar death rate was observed in a previous trial of people with TB who smoke.¹⁸ The probability of death was significantly higher in the usual care group compared with the mHealth group. This measurable mortality reduction within a short follow-up period of 6 months is striking and should be confirmed in future studies. The association between smoking and TB mortality is well established³⁴; hence, a higher abstinence rate (as observed here) may have prevented deaths in the mHealth group compared with the usual care group. A meta-analysis of 16 RCTs of psychosocial interventions in medical settings reported improved patient

survival compared with controls, specifically when psychosocial interventions explicitly promoted healthy behaviors.³⁵

Limitations

The trial had several limitations. First, participants were almost entirely male (>96%). This is not surprising, given that in Bangladesh and Pakistan, 97.8% and 91.4%, respectively, of all adult individuals who smoke are men. Second, the comparator group received usual care; there was no attention control. Third, a substantial proportion of participants had no formal education, and it could not be confirmed if they understood the text messages. Fourth, mHealth fidelity was monitored by calling participants regularly, which could also act as an intervention; hence, the results must be interpreted in light of this element. Fifth, while measuring levels of breath CO is an established method to validate self-reported abstinence, it can only verify short-term (24-48 hours) abstinence due to its short half-life (5-6 hours).³⁶ Sixth, we did not follow participants beyond their TB treatment, and therefore we cannot confirm if abstinence was sustained; however, a previous smoking cessation trial in people with TB reported that 3 of 4 people were still abstinent at 1 year.¹⁸ Additionally, the observed intracluster correlation coefficient (0.19) was substantially higher than the value (0.02) assumed for sample size calculation. However, the intervention still produced a statistically significant effect, indicating robustness.

Conclusions

Among people with TB who smoke tobacco, an mHealth intervention was effective in promoting continuous abstinence. mHealth is a feasible, effective, and potentially scalable means of supporting tobacco cessation.

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